

Madhusoodhan Tirunangur Girinarayanan

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SUMMARY

ML Engineer completing M.S. Computer Science at Illinois Tech (May 2026), with production experience building LLM-powered NLP pipelines and large-scale data infrastructure on AWS SageMaker. Designed and shipped end-to-end language understanding systems - including a multilingual clinical NLP pipeline achieving 18.3% WER and a RAG-driven feature retrieval system across 132M+ records - with hands-on depth in fine-tuning, RAG, and LLM orchestration using PyTorch and the Claude API.

EXPERIENCE

Czero LLC | Machine Learning Engineer - Chicago, IL

Jan 2026 - May 2026

- Architected an end-to-end data pipeline on AWS SageMaker processing 132M+ industrial records, designing scalable infrastructure for high-throughput feature extraction, model training, and low-latency model serving in production.
- Built a RAG-inspired contextual retrieval pipeline on high-dimensional datasets to surface relevant historical signals for downstream models, improving inference accuracy and enriching feature representations.
- Engineered sequential prediction model infrastructure with GPU-accelerated inference, optimizing latency-sensitive workloads against high-throughput data streams in production environments.
- Monitored model performance and data quality continuously in production, applying statistical imputation (MICE) at scale to maintain pipeline integrity and reduce model degradation caused by missing signals.
- Collaborated cross-functionally with manufacturing and engineering teams to align model outputs with business requirements, communicating model behavior to non-technical stakeholders.

UCAL Ltd. | Developer - Chennai, India

Jun 2022 - Jul 2025

TPMS (Tire Pressure Monitoring System)

- Built an Android HMI application integrating TPMS sensors across two-wheelers to heavy trucks via Bluetooth, enabling real-time tire pressure and temperature monitoring with automated analysis and predictive tire replacement alerts.
- Designed an anomaly detection and alerting system for electrical equipment failures, reducing unplanned downtime by 40% through proactive threshold monitoring and automated maintenance notifications.
- Engineered a battery and sensor power management system that dynamically drives both phone and TPMS hardware into rest mode when idle, minimizing power draw across long-haul routes while maintaining instant wake-on-trigger responsiveness.

CDI (Capacitive Discharge Ignition)

- Collaborated on R&D of a Capacitive Discharge Ignition system for two-wheelers, using Spectra to capture and analyze ignition pulse waveforms and measure firing frequency to characterize component timing behavior.
- Programmed and interfaced with a Renesas microcontroller module embedded in the CDI unit, reconstructing the optimal ignition waveform from validated reference measurements and achieving a 15% improvement in firing performance over the baseline CDI design.

PROJECTS

ClinScribe - Multilingual AI Clinical Scribe | [GitHub](#)

- Led ML infrastructure design for a 5-member team, architecting a full language understanding pipeline from raw Hindi audio ingestion through clinical entity extraction, medical code mapping, and FHIR R4 structured knowledge write-back - enabling downstream teams to consume enriched content at scale.
- Built an end-to-end NLP enrichment pipeline using Claude API for LLM-driven entity extraction, SNOMED-CT/ICD-10 runtime code lookup, and structured knowledge population via FHIR R4 writes to OpenMRS via Docker, with full clinician review UI in FastAPI.
- Designed a privacy-preserving language understanding system running Whisper large-v3 locally for Hindi clinical audio transcription (PHI - no external API calls), achieving 18.3% WER below the 20% clinical usability threshold.
- Developed a benchmarking evaluation framework comparing language model performance across 7 clinical scenarios and 2 languages (Hindi 18.3% WER vs. Swahili 21.8%), establishing data-driven selection criteria for production deployment.
- Designed offline deployment architecture using vLLM for local LLM inference, enabling fully network-less language understanding in low-connectivity healthcare settings.

Agastya - AI Medication Intelligence System | *Hackathon Winner, ScarlettHacks* | [GitHub](#)

- Architected a full-stack LLM-powered content understanding pipeline integrating Claude Vision API to extract and interpret handwritten prescription text across 4 regional scripts (Kannada, Hindi, Tamil, English), applying language understanding to surface real-time drug interaction knowledge - built end-to-end in 24 hours.
- Employed RAG-inspired retrieval to cross-reference user medication history against structured drug knowledge bases, personalizing recommendations based on individual user context.
- Built a multilingual serving layer with in-memory translation caching, Web Speech API voice output in 5 languages, and a Firebase + IndexedDB backend enabling zero-latency repeated interactions.

Nexus - Locally-Run Personal AI Agent | [GitHub](#)

- Built a Claude API tool-use orchestration loop (40+ tools, up to 10 rounds per request) with Anthropic prompt caching, applying structured prompting techniques to specialize LLM behavior for domain-specific language understanding tasks.
- Architected a multi-adaptor LLM serving system with a serialized request queue and exponential backoff, achieving production-grade reliability across concurrent CLI and WhatsApp inputs.
- Designed a three-tier trust model with behavioral anomaly detection, SQLite-persisted long-term memory, and confirmation gates for sensitive operations - running fully locally on Linux.

EDUCATION

Illinois Institute of Technology | M.S. Computer Science | Chicago, IL May 2026

Rajalakshmi Engineering College | B.Tech, Information Technology | Chennai, India May 2024

SKILLS

ML Infrastructure & Platforms: AWS SageMaker, AWS S3, AWS Athena, Apache Spark, Docker, Kubernetes, Apache Airflow, Cloud-native deployment, MLOps pipelines, GPU inference (NVIDIA RTX), Feature engineering, Model serving

AI & LLM Systems: Claude API (vision + text), Claude Code, OpenAI API, LangChain, RAG, Fine-tuning, Prompt Engineering, Multi-agent orchestration, Whisper large-v3, vLLM, FHIR R4

NLP & Content Understanding: Language Understanding, Named Entity Recognition, Text Classification, SNOMED-CT/ICD-10 mapping, Multilingual ASR, Content Enrichment Pipelines

ML Frameworks & Architectures: PyTorch, TensorFlow, Keras, Scikit-Learn, YOLOv3, ResNet, U-Net, Transformer models, Time-series forecasting

Programming & Backend: Python, Scala, Flask, FastAPI, REST APIs, PostgreSQL, Git, Firebase, Go, C++, C, C#

Frontend: TypeScript, React, JavaScript, Web Speech API

CERTIFICATIONS & AWARDS

Machine Learning (Supervised Learning) - Stanford Online / Coursera

HIPAA Certification

Hackathon Winner - Agastya AI Medication Intelligence System - ScarlettHacks